Care Access Navigator

Incident Response 1 — Task 4: Digital Evidence Management

Live data collection, Volatility memory analysis, disk acquisition, chain of custody and a macOS + Parrot OS timeline for IR1-2026-002

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Case	**IR1-2026-002**, SEV-1: a mover's stale clinical access was used to stage and upload the synthetic record PT-1003 (detected in Task 2, contained in Task 3)
Evidence source	parrot-vm (Debian 12 VM, Parrot OS stand-in; kernel 6.1.0-54-cloud-amd64) in its contained state, plus the Task 3 acquisitions; macOS host DAEDMAC1009
Workstation	ir-forensics (Parrot forensic toolset): dc3dd 7.2.646, The Sleuth Kit 4.11.1, Volatility 3 Framework 2.7.0, TShark 4.0.17, ewf-tools
Run	19:37:19–19:39:48 UTC, 2026-11-02; 66 files in the evidence index; 13 evidence items in the chain of custody

Summary

Every Task 3 acquisition was re-hashed at intake (29 of 29 OK). Live data was then collected from the contained host: 24 commands in order of volatility, written to RAM only, each with a logged SHA-256 (24/24 re-verified). Volatility 3 analysed the Task 3 RAM image with a symbol table built for this exact kernel. It found the uploader (PID 1534) with the attacker's SSH session in its environment, an open socket stuck in SYN_SENT to the drop site, the patient record in its heap, and the kernel's own log naming it as the source of the firewall drops. No rootkit indicators were found. The disk was seized at the hypervisor and imaged with dc3dd from a read-only source. The source, dc3dd input and image hashes are identical, and The Sleuth Kit recovered the three planted files byte-identical to the live copies. A chain-of-custody register tracks 13 items through 54 custody events. The super-timeline merges 187 events from 11 macOS and Parrot sources. Every macOS SSH session matches a Parrot sshd login by key fingerprint to within a second.

1. How this report meets the Task 4 requirements

Requirement	Where	Evidence
Live data collection: system state	§3	live_response/01_clock, 02_system, 14–16
Live data collection: running process information	§3	live_response/03_processes, 04_process_exe_sha256, 05_open_files
Live data collection: network connection data	§3	live_response/07_sockets, 08_listening, 09_interfaces, 10_firewall
Basic memory analysis using Volatility	§4	memory/vol_*.txt (14 plugins), uploader_region_search.txt

Requirement	Where	Evidence
Disk acquisition with Parrot imaging tools, proper procedure	§5	disk/dc3dd_acquisition.log, mmls, fsstat, fls, istat
Chain of custody for all collected evidence	§6	coc/evidence_register.csv, coc/custody_log.csv
Timeline with macOS and Parrot OS logs, with analysis	§7	timeline/supertimeline, key_events, session_correlation

2. Evidence handling procedure

The procedure follows NIST SP 800-86's four phases (collection, examination, analysis, reporting) and RFC 3227's order of volatility. These rules applied to every item:

Rule	How it was applied here
Most volatile first	Task 3 took RAM and packets before files. Here, clock → processes → network → kernel → accounts → file system
Do not write to the suspect disk	Live-response output goes to /dev/shm (RAM) and is streamed off; the disk is imaged from a hypervisor export, not from inside the guest
Write-protect originals	Seized disk mounted read-only into the workstation (write attempt refused, §5); large evidence files set read-only (`chmod a-w`)
Hash at acquisition, verify at every hand-off	SHA-256 at source and destination; intake re-verification; hash before/after each examination
Examine copies, keep originals	Sleuth Kit runs on the dc3dd working copy (E12), never on the seized original (E11)
Document everything	Every command is in transcripts/ with UTC times; each live-response output is tied to its command in 00_collection_log.tsv
Clock reference	The suspect host's clock is recorded against the workstation clock (§3.2) before any timestamp is trusted

transcripts/01_intake_verification.txt - intake check of the Task 3 evidence

```
# re-hash every Task 3 evidence item against its acquisition hash (task3 EVIDENCE-INDEX-SHA256.txt)
total 2123208
-r--------  1 Adult  staff  1074005971 Nov  3 14:32 IR1-2026-002_parrot-vm.mem.elf
-r--r--r--  1 Adult  staff    11973688 Nov  3 14:32 IR1-2026-002_syncd.core
repo items:  27 OK, 0 not OK
large items: 2 OK, 0 not OK
IR1-2026-002_parrot-vm.mem.elf: OK
IR1-2026-002_syncd.core: OK
```

3. Live data collection from Parrot OS

`lab/ir_live_response.sh` is a small, readable live-response collector (SHA-256 of the script is in transcript 02). It was piped to the contained host over the forensic SSH channel and run as root. That is the only path left open after containment. Each command's output went to RAM. The collector logged the command line, UTC start time, exit code and the output's SHA-256 next to it. The workstation then pulled the set as a tar and re-hashed every file against that log.

#	Output	Category	UTC (host clock)	Exit	SHA-256
1	01_clock.txt	System state	19:37:15.730	0	bf5d8f4fd74e6ec5…
2	02_system.txt	System state	19:37:16.566	0	ac477e92418469ec…
3	03_processes.txt	Processes	19:37:16.772	0	f27e842b39dc254d…
4	04_process_exe_sha256.txt	Processes	19:37:16.990	0	373b38e0bc885714…
5	05_open_files.txt	Processes	19:37:21.244	0	b7504b32ee4fe303…
6	06_logged_in.txt	System state	19:37:21.774	0	0fe81e4e60dab3c8…
7	07_sockets.txt	Network	19:37:21.930	0	2b33344eaa500370…
8	08_listening.txt	Network	19:37:22.196	0	a3e178e7b509d896…
9	09_interfaces.txt	Network	19:37:22.366	0	4faebcf351180d0e…
10	10_firewall.txt	Network	19:37:22.575	0	a204d992a62c3743…
11	11_dns_hosts.txt	Network	19:37:22.764	0	f1d272f1b721bd55…
12	12_kernel_modules.txt	Kernel/services	19:37:22.884	0	6c259643e1afd5fc…
13	13_kernel_log.txt	Kernel/services	19:37:23.020	0	9cdc73380dde6ffc…
14	14_services.txt	Kernel/services	19:37:23.183	0	c60e63330200dc1a…
15	15_timers_cron.txt	Kernel/services	19:37:23.344	0	15a14030a41165f5…
16	16_mounts_disks.txt	System state	19:37:23.699	0	511113ca2f128b86…
17	17_accounts.txt	Accounts	19:37:23.973	0	8d8e1cb55eca11de…
18	18_sudoers.txt	Accounts	19:37:24.538	0	57e69d2aaf4a5613…
19	19_ssh_keys.txt	Accounts	19:37:24.703	0	7234b6aae85ed0c9…
20	20_lastlog.txt	Accounts	19:37:25.071	0	d227418b4356b3c7…
21	21_recent_files.txt	File system	19:37:25.244	0	dbc46ac7c74ee77c…
22	22_hidden_dirs.txt	File system	19:37:25.507	0	a87753adc849440f…

#	Output	Category	UTC (host clock)	Exit	SHA-256
23	23_suid_sgid.txt	File system	19:37:25.647	0	15474608a0c119a8...
24	24_shell_history.txt	File system	19:37:28.406	0	0adbcb6e9780d6f6...

3.1 What the live data shows

17_accounts.txt - containment state of the accounts

```
root:x:0:0:root:/root:/bin/bash
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
ir-analyst:x:1000:1000:Incident Response analyst (forensic access):/home/ir-analyst:/bin/bash
it_admin:x:1001:1004:IT Administrator (lab):/home/it_admin:/bin/bash
test.clinician:x:1002:1005:Sathya Hasi (Physician):/home/test.clinician:/bin/bash
test.nurse:x:1003:1006:Tara Jones (Billing Specialist):/home/test.nurse:/bin/bash
test.billing:x:1004:1007:Jeevan Smith (Billing Specialist):/home/test.billing:/bin/bash

sudo:x:27:it_admin
clinical:x:1001:test.clinician
billing:x:1002:test.nurse,test.billing
disabled-users:x:1003:test.billing,test.nurse
test.billing:x:1007:

nobody L 2026-10-26 0 99999 7 -1
ir-analyst L 2026-11-02 0 99999 7 -1
it_admin L 2026-11-03 0 99999 7 -1
test.clinician P 2026-11-03 0 99999 7 -1
test.nurse L 2026-11-03 0 99999 7 -1
test.billing L 2026-11-03 0 99999 7 -1
```

21_recent_files.txt (filtered) - files written during the incident and the response

```
2026-11-03 19:20:55.3520000000  root    600 114 /etc/netplan/60-ir-lab.yaml
2026-11-03 19:31:08.7336674870  test.nurse  644 127 /home/test.nurse/.cache/.sync/pt1003.txt
2026-11-03 19:31:09.3816872490  test.nurse  700 827 /home/test.nurse/.cache/.sync/syncd.py
2026-11-03 19:31:11.8297618440  test.nurse  600 280 /var/spool/cron/crontabs/test.nurse
2026-11-03 19:34:21.1736889160  root    644 46  /etc/ssh/sshd_config.d/00-ir-containment.conf
```

07_sockets.txt - only the responder's SSH session remains

```
Netid State  Recv-Q Send-Q Local Address:Port  Peer Address:Port Process

udp   UNCONN 0      0            0.0.0.0:5355       0.0.0.0:*
    users:(("systemd-resolve",pid=313,fd=11))
udp   UNCONN 0      0         127.0.0.54:53         0.0.0.0:*
    users:(("systemd-resolve",pid=313,fd=19))
udp   UNCONN 0      0      127.0.0.53%lo:53         0.0.0.0:*
    users:(("systemd-resolve",pid=313,fd=17))
udp   UNCONN 0      0               [::]:5355          [::]:*
    users:(("systemd-resolve",pid=313,fd=13))
tcp   LISTEN 0      20         127.0.0.1:25         0.0.0.0:*     users:(("exim4",pid=647,fd=4))

tcp   LISTEN 0      4096      127.0.0.54:53         0.0.0.0:*
    users:(("systemd-resolve",pid=313,fd=20))
tcp   LISTEN 0      4096         0.0.0.0:5355       0.0.0.0:*
    users:(("systemd-resolve",pid=313,fd=12))
```

- **Processes** (`03_processes.txt`): no `syncd.py` and no process owned by test.nurse. The uploader was killed in Task 3 C7, so its last running state comes from memory (§4).

- **Network** (07–10): one live TCP session (the responder, from 10.78.50.5 on the quarantine NIC). sshd listens only on 10.0.3.15, and the default-DROP ruleset from Task 3 is still loaded.
- **Persistence** (`15_timers_cron.txt`): the test.nurse crontab is still on disk (kept as evidence), but cron is masked, so it cannot run.

3.2 Clock check

The host reported **2026-11-03 19:37:16 UTC** while the workstation read **2026-11-03 19:37:22 UTC**: the host was **6 s slow**. Its uptime was 20.5 minutes, although about 22 minutes of real time had passed since boot. The emulated guest clock runs slow, and isolation cut it off from NTP, so nothing corrects it while it is contained; the gap grows with time. The timestamps that matter are not affected. Before isolation the host was NTP-synced: its sshd log agrees with the Mac's SSH client to within a second for every session (§7.2). Host times written after isolation are used for order only, not wall-clock time, and the live-response times above are host-clock values.

> **Why it matters:** In an earlier test run the contained guest was left for over an hour and drifted 75 minutes. Without a clock reference its live-response times would have looked 75 minutes older than they were. Recording host time against a trusted reference before any timestamp is used is a required step, not a formality.

4. Memory analysis with Volatility 3

Volatility needs a symbol table (ISF) for the exact kernel in the image. None was published for Debian's 6.1.0-54-cloud-amd64, so it was built with the Volatility Foundation's `dwarf2json` v0.9.0 from Debian's own debug package (`linux-image-6.1.0-54-cloud-amd64-dbg_6.1.190-1`). It is stored in `lab/symbols/`. The image is E01, the 1 GiB RAM dump taken by the hypervisor in Task 3 before the uploader was killed.

Image integrity check	SHA-256 of E01
Before analysis	`7010d346a57e2818d14b61d8ba9dd780258f73cf3c3f7bc77a4c5789d4fe9a0c`
After analysis	`7010d346a57e2818d14b61d8ba9dd780258f73cf3c3f7bc77a4c5789d4fe9a0c` **UNCHANGED**

linux.psaux - the uploader (and the EHR portal) as they were in memory

```
PID PPID    COMM    ARGS
375 1   cron    /usr/sbin/cron -f
390 1   unattended-upgr /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown
    --wait-for-signal
811 1   sshd    sshd: /usr/sbin/ss
852 1   python3 /usr/bin/python3 -m http.server 8080 --directory /srv/ehr/portal
1534    1   python3 python3 /home/test.nurse/.cache/.sync/syncd.py
```

linux.envars --pid 1534 - the process carries the SSH session it came from

```
PID       PPID      COMM      KEY       VALUE
1534      1         python3   PWD       /home/test.nurse
1534      1         python3   HOME      /home/test.nurse
1534      1         python3   SSH_CONNECTION      10.78.10.1 62768 10.0.2.15 22
1534      1         python3   USER      test.nurse
1534      1         python3   SSH_CLIENT          10.78.10.1 62768 22
```

linux.sockstat - the uploader's socket to the drop site

```
NetNS     Pid       FD        Sock Offset         Family    Type      Proto     Source Addr
    Source Port
4026531840          1534      3         0x8b1782486c00      AF_INET   STREAM    TCP       10.0.2.15
    55446     10.78.99
```

linux.lsof --pid 1534

```
PID       Process   FD        Path
1534      python3   0         /dev/null
1534      python3   1         /dev/null
1534      python3   2         /dev/null
1534      python3   3         socket:[20614]
```

linux.kmsg - kernel log in RAM: 13 IR-BLOCK lines

```
kern    warn    915.744987      Task(1534)      IR-BLOCK IR1-2026-002 IN= OUT=enp0s1 SRC=10.0.2.15
    DST=10.78.99.66 LEN
kern    warn    916.802432      CPU(1)  IR-BLOCK IR1-2026-002 IN= OUT=enp0s1 SRC=10.0.2.15
    DST=10.78.99.66 LEN=60 TOS=
kern    warn    918.846273      CPU(1)  IR-BLOCK IR1-2026-002 IN= OUT=enp0s1 SRC=10.0.2.15
    DST=10.78.99.66 LEN=60 TOS=
```

linux.proc.Maps --pid 1534 --dump, then grep - patient record in the heap

```
regions dumped: 85
region holding MRN SYN-0003: pid.1534.vma.0x7f0123f2c000-0x7f012418e000.dmp
SYNTHETIC RECORD PT-1003
```

4.1 Findings and integrity checks

Plugin	Result	Meaning
pslist / pstree / psaux	90 tasks; PID 1534 = python3 syncd.py, parent 1	Detached (setsid) process: PPID 1, as planted
envars	SSH_CONNECTION=10.78.10.1 … 10.0.2.15 22	Started from Tara's SSH session (NAT address of the Mac)
sockstat	TCP 10.0.2.15 → 10.78.99.66:8080 SYN_SENT	Still trying to upload; held by the C3 firewall drop
lsof	fd 0–2 /dev/null, fd 3 socket	Fully detached from any terminal
kmsg	13 IR-BLOCK lines, first attributed to Task(1534)	Memory confirms what kern.log said (Task 3 §5.2)
proc.Maps + dump	Record text and MRN in a heap region	Proves the process held the ePHI, not just a file path
bash	no entries	Commands were sent as SSH exec, not typed into an interactive shell, so the trail is in auditd (§7)
check_modules / lsmod	no hidden modules; 52 listed	No kernel rootkit
check_syscall	430 entries, 0 without a kernel symbol	**Clean**: no syscall hooks
malfind	no injected executable regions	**Clean**

Validation: Cross-check with the live data: Task 3's `ps` listed the uploader as PID 1534, started 19:31 UTC by test.nurse. Volatility, reading raw memory without the guest's tools, finds the same PID, command line and owner. A tampered ps would not survive this comparison.

Configuration point: The first Volatility attempt returned no processes. The guest had booted with 5-level paging (QEMU's full CPU model), which Volatility 3 2.7.0 does not translate. The lab therefore runs the guest with `-cpu max,la57=off` (Task 1 §7), and this image comes from it.

5. Disk acquisition with Parrot OS imaging tools

The suspect is a virtual machine, so the disk was acquired the way a VM disk is acquired: seized at the hypervisor from a fixed point in time, then imaged and verified on the forensic workstation. Imaging from inside the running guest would read a live, changing file system and would have to write the image through the suspect's own kernel.

#	Step	Command / tool	Result
1	Seize: export the disk as at snapshot ir-contained	`qemu-img convert -U -l snapshot.name=ir-contained -O raw` (= VBoxManage clonemedium --format RAW)	E11, 8 GiB, sha256 `fa80be62ad1c0568…`
2	Write-block the source	E11 mounted read-only into ir-forensics at /seized	write attempt refused: *Read-only file system*
3	Image with hashing and log	`dc3dd if=/seized/... of=...vda.dd hash=sha256 log=dc3dd_acquisition.log`	16,777,216 sectors in = out; 18.1087 s, 452 M/s
4	Verify	sha256sum on the image vs dc3dd's input hash vs step 1	**all three equal**
5	Protect	image and seized file set read-only; hashes into the evidence index	E12 = working copy
6	Examine (copy only)	mmls, fsstat, fls, ifind, icat, istat, mactime (The Sleuth Kit)	§5.1

disk/dc3dd_acquisition.log

```
dc3dd 7.2.646 started at 2026-11-03 19:39:04 +0000
compiled options:
command line: dc3dd if=/seized/IR1-2026-002_parrot-vm_seized.raw
    of=/evid/IR1-2026-002/IR1-2026-002_vda.dd hash=sha256
    log=/ir/task4_evidence_management/evidence/disk/dc3dd_acquisition.log
sector size: 512 bytes (assumed)
  8589934592 bytes ( 8 G ) copied ( 100% ), 18.1087 s, 452 M/s
input results for file `/seized/IR1-2026-002_parrot-vm_seized.raw':
   16777216 sectors in
   fa80be62ad1c05688c43f665c72e49be438ab46bb7232bb19a5ba7a01e0b7cbb (sha256)
output results for file `/evid/IR1-2026-002/IR1-2026-002_vda.dd':
   16777216 sectors out
dc3dd completed at 2026-11-03 19:39:22 +0000
```

5.1 Examination of the image

mmls - GPT; the root file system is the largest partition, at sector 262144

```
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  013       0000002048   0000008191   0000006144
005:  014       0000008192   0000262143   0000253952
006:  000       0000262144   0016777182   0016515039
007:  -------   0016777183   0016777215   0000000033   Unallocated
```

fsstat (excerpt)

```
$ fsstat -o 262144 /evid/IR1-2026-002/IR1-2026-002_vda.dd | head
File System Type: Ext4
Volume Name:
Volume ID: 6110e0769bc0698a5746e914550aab0e
Last Written at: 2026-10-26 04:46:07 (UTC)
Last Checked at: 2026-10-26 04:42:26 (UTC)
Last Mounted at: 2026-11-03 19:16:45 (UTC)
Compat Features: Journal, Ext Attributes, Resize Inode, Dir Index
Journal ID: 00
Journal Inode: 8
```

fls -r -p (incident paths) - planted files, quarantined key, containment config

```
d/d 29039:  home/test.nurse
r/r 31393:  home/test.nurse/.bashrc
r/r 31397:  home/test.nurse/.profile
r/r 31403:  home/test.nurse/.bash_logout
d/d 31426:  home/test.nurse/.ssh
d/d 32023:  home/test.nurse/.cache
d/d 32024:  home/test.nurse/.cache/.sync
r/r 32025:  home/test.nurse/.cache/.sync/pt1003.txt
r/r 32027:  home/test.nurse/.cache/.sync/syncd.py
r/r 31406:  var/spool/cron/crontabs/test.nurse
r/r 32013:  etc/ssh/sshd_config.d/00-ir-containment.conf
r/r 32028:  root/iptables.pre-containment
r/r 32030:  root/iptables.containment
d/d 31437:  root/ir-quarantine
r/r 31441:  root/ir-quarantine/test.nurse.authorized_keys
r/r 31464:  root/ir-quarantine/test.billing.authorized_keys
```

File recovered from the image (icat)	Inode	Image SHA-256 = Task 3 live copy?
`/home/test.nurse/.cache/.sync/syncd.py`	32027	**MATCH**
`/home/test.nurse/.cache/.sync/pt1003.txt`	32025	**MATCH**
`/var/spool/cron/crontabs/test.nurse`	31406	**MATCH**

istat syncd.py - owner uid 1003 (test.nurse), mode 0700, created 19:31:09 UTC

```
inode: 32027
Allocated
Group: 3
Generation Id: 3498220599
uid / gid: 1003 / 1006
mode: rrwx------
Flags: Extents,
size: 827
num of links: 1

Inode Times:
Accessed:   2026-11-03 19:31:09.413688225 (UTC)
```

6. Chain of custody

`analysis/chain_of_custody.py` builds the register and the custody log from the evidence itself. Times come from the run markers, acquisition hashes from the transcripts, and every item is re-hashed when the log is built. Each item has an ID (E01–E13), a source, a method, an acquirer, a storage location and two hashes: at acquisition and now.

ID	Item	Acquired (UTC)	Stored	SHA-256 at acquisition	Now
E01	Full RAM image of parrot-vm (1 GiB ELF core)	19:32:32	~/ir1-vm-lab/evidence (outside git)	`7010d346a57e…`	**OK**
E02	Process memory core of uploader syncd.py	19:32:32	~/ir1-vm-lab/evidence (outside git)	`1d4ccb577d0d…`	**OK**
E03	Live packet capture, host side (25 s, host 10.78.99.66)	19:31:28	task3/network/IR1-2026-002_live_tcpdump.pcap	`ef0c2bf5c2d3…`	**OK**
E04	Hypervisor capture of adapter 1 (clinical), whole run	19:35:47	task3/network/hypervisor_nic0_clinical.pcap	`3967907c437e…`	**OK**
E05	Hypervisor capture of adapter 2 (quarantine)	19:35:47	task3/network/hypervisor_nic1_quarantine.pcap	`f1d996df10dc…`	**OK**
E06	Volatile state set (processes, connections, suspect process, persisten	19:31:28	task3/volatile/processes.txt	`af5b79881da6…`	**OK**
E07	File-system artifacts tar (12 files incl. staged record, uploader, cro	19:32:47	task3/filesystem/IR1-2026-002_artifacts.tar	`871eb2e5c6b2…`	**OK**
E08	Drop-site upload log	19:35:47	task3/exfil-sink/uploads.jsonl	`9b60c46749c9…`	**OK**
E09	macOS SSH client log	19:35:47	task2_log_analysis/evidence/macos/ssh_client.log	`3b023d79493b…`	**OK**
E13	Wazuh alerts (SIEM record of the incident, incl. rule 100221 that trig	19:35:47	task2_log_analysis/evidence/wazuh/alerts.json	`8f44f43e075a…`	**OK**
E10	Live-response collection (24 command outputs + collection log)	19:37:22	task4/live_response/00_collection_log.tsv	`fd236a6a68c6…`	**OK**
E11	Seized virtual disk, raw (state at snapshot ir-contained)	19:38:42	~/ir1-vm-lab/evidence (outside git)	`fa80be62ad1c…`	**OK**
E12	Forensic disk image (dc3dd) of E11	19:39:04	~/ir1-vm-lab/evidence (outside git)	`fa80be62ad1c…`	**OK**

The custody log has 54 events: 13 acquired, 13 stored, 9 verified (intake to task 4), 2 examined, 13 verified (report build), 4 transfer. The full history of the disk image (E12) shows each hand-off and its hash check:

UTC	Event	From → To	Check
19:39:04	ACQUIRED	E11 (read-only mount) → ir-forensics	sha256 recorded

UTC	Event	From → To	Check
19:39:04	TRANSFER	E11 (read-only mount) → ir-forensics /bigev	dc3dd input hash = E11 hash: MATCH
19:39:04	STORED	ir-forensics → IR1-2026-002_vda.dd	file set read-only
19:39:39	EXAMINED	IR1-2026-002_vda.dd → The Sleuth Kit (read-only)	image not modified
19:39:40	VERIFIED (report build)	IR1-2026-002_vda.dd → IR1-2026-002_vda.dd	MATCH

Custody form field	Value for every item in this case
Case / incident	IR1-2026-002 (SEV-1), Care Access Navigator, HealthLine Regional Medical Center (HRMC) (lab)
Collected by / custodian	A. Chennamaneni, IR analyst (account ir-analyst); custodian of the case store
Storage	Repository evidence/ folders (git history = tamper-evident log of changes); images > 100 MB in ~/ir1-vm-lab/evidence on the analyst's FileVault-encrypted macOS volume (fdesetup status: On), set read-only
Access	Images: analyst's local account only. The course repository is **public** on GitHub. That is acceptable here only because every record is synthetic; real ePHI evidence must never go into a code repository. Any copy for a third party (Privacy Officer, legal) is a new custody event: who, when, why, hash before and after
Retention	Kept until the HIPAA breach-risk assessment for PT-1003 is closed, then per the record-retention schedule (Compliance 1 Task 4)

7. Timeline: macOS and Parrot OS

`analysis/build_timeline.py` merges every timestamped source into one UTC table (`timeline/supertimeline_IR1-2026-002.csv`, 187 events). Each source was normalised to UTC: the Mac's -05:00 local time was converted, auditd epoch seconds were converted, and mactime ran with -z UTC.

Host	Source	Events
DAEDMAC1009	`ssh_client.log`	16
IR	`task3 markers.csv`	11
IR	`task4 steps.csv`	8
exfil-sink	`uploads.jsonl`	10
ir-hv	`nic0 pcap`	10
parrot-vm	`audit.log`	38
parrot-vm	`auth.log`	45
parrot-vm	`ehr-audit.log`	2
parrot-vm	`kern.log`	15

Host	Source	Events
parrot-vm	`mactime`	11
wazuh-manager	`alerts.json`	21

> **Parser bug found and fixed:** auditd restarts its record serial number at every boot, and the golden image's audit.log also holds the image-build boot. Joining records on the serial alone attached commands from the lab boot to timestamps from the image-build boot. Records are now joined on time + serial, and the timeline is limited to this lab run.

7.1 Key events

UTC	Milestone	Source
19:31:06.000	Tara's first SSH session from the Mac authenticated	`ssh_client.log`
19:31:06.791	Parrot sshd accepts test.nurse (same key fingerprint)	`auth.log`
19:31:08.718	PT-1003 viewed through stale clinical group	`ehr-audit.log`
19:31:08.733	auditd: patient file read	`audit.log`
19:31:08.000	Staged copy written (file system)	`mactime`
19:31:09.000	Uploader script written (file system)	`mactime`
19:31:09.985	Uploader started (auditd)	`audit.log`
19:31:11.813	Persistence: crontab written (auditd)	`audit.log`
19:31:10.550	First upload received by drop site	`uploads.jsonl`
19:31:10.008	Wazuh alert 100221 (detection, SEV-1)	`alerts.json`
19:31:12.000	Triage confirms disclosure in progress	`task3 markers.csv`
19:31:59.000	Egress blocked (C3)	`task3 markers.csv`
19:31:56.803	Last upload received by drop site	`uploads.jsonl`
19:32:00.788	First firewall drop logged	`kern.log`
19:32:32.000	Memory preserved (C4)	`task3 markers.csv`
19:32:52.000	Host isolated (C6)	`task3 markers.csv`
19:34:16.475	Account locked + clinical group removed	`auth.log`
19:35:47.000	Contained	`task3 markers.csv`

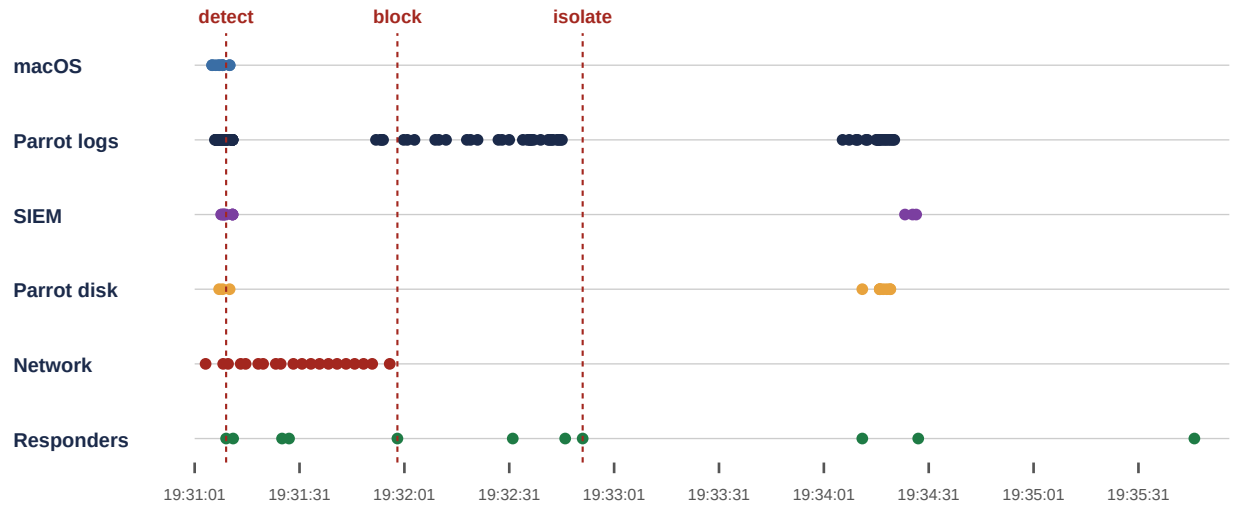

Figure 1. Super-timeline, 19:31:01–19:35:57 UTC. Each dot is one event. Network dots stop at the block line; Parrot dots after it are the firewall logging dropped attempts.

7.2 Cross-host correlation (macOS ↔ Parrot OS)

The Mac and parrot-vm see different addresses for the same session, because the Mac connects to 127.0.0.1:2223 and the VM sees the Docker NAT address. They were therefore joined on two independent keys: time, and the SSH key fingerprint the Mac offered against the one Parrot's sshd accepted.

Mac ssh PID	Mac authenticated	Parrot sshd accepted	Δ s	Key fingerprint
48924	19:29:50	19:29:50.404	+0.40	**MATCH** `SHA256:XpRA2t9M6w0xG...`
48981	19:30:14	19:30:14.029	+0.03	**MATCH** `SHA256:XpRA2t9M6w0xG...`
49100	19:31:06	19:31:06.791	+0.79	**MATCH** `SHA256:XpRA2t9M6w0xG...`
49105	19:31:08	19:31:08.275	+0.28	**MATCH** `SHA256:XpRA2t9M6w0xG...`
49109	19:31:09	19:31:09.069	+0.07	**MATCH** `SHA256:XpRA2t9M6w0xG...`
49113	19:31:09	19:31:09.069	+0.07	**MATCH** `SHA256:XpRA2t9M6w0xG...`
49117	19:31:11	19:31:11.488	+0.49	**MATCH** `SHA256:XpRA2t9M6w0xG...`

7.3 Analysis

- **Intent was immediate.** 3 s passed from first login to viewing PT-1003, and 1.8 s more to the first upload. Staging, the uploader and the cron job were all in place within 5 s, an order of operations that only makes sense if planned.
- **Every layer saw the read.** The EHR audit trail (application), auditd (kernel watch on the file), the file system (the staged copy's birth time) and memory (the heap) all place PT-1003 in test.nurse's hands at 19:31:08–19:31:09 UTC.
- **Containment closed the window.** The last upload reached the drop site at 19:31:56. Egress was blocked at 19:31:59, and the first logged drop came at 19:32:00. The block took effect at the

uploader's next 5-second cycle.

- **Root cause visible in the timeline:** the mover change (Nurse → Billing) added billing but never removed clinical. The account change that finally removed it is the C7 entry at 19:34:16, after the incident. That is the gap Task 6 fixes in the JML process.

8. Limitations

- Parrot OS and VirtualBox are substituted by a Debian 12 VM under QEMU (see Task 3 §9). All tools used are in Parrot's standard forensic set, and every hypervisor step has a VirtualBox equivalent.
- The macOS side is the SSH client log written on the Mac. The macOS unified log needs admin rights to read on this managed Mac (Task 2 §5).
- The image is raw (dd), not E01. ewfacquire is installed; raw was chosen so the Sleuth Kit and hash checks run on a plain bit-for-bit copy.

Appendix A — Evidence index

evidence/EVIDENCE-INDEX-SHA256.txt (hashes truncated here)

```
6abccce9cc31bcee...  ./coc/custody_log.csv
e383117420bca5ff...  ./coc/evidence_register.csv
a28082dddd769d76...  ./disk/bodyfile_incident.txt
51edb1db5aca8af1...  ./disk/dc3dd_acquisition.log
cd18251750b21171...  ./disk/fls_incident_paths.txt
ac8819c040852315...  ./disk/fsstat.txt
142677fab5650fb8...  ./disk/image_auth.log
21f89e60217f451a...  ./disk/image_ehr-audit.log
1366d6c69e5ab6ac...  ./disk/istat_syncd.txt
0328015da7e8101e...  ./disk/mactime_incident.csv
7577bc62bb79dc2d...  ./disk/mmls.txt
557656cc33b5f946...  ./disk/recovered_vs_task3.txt
09dc316754fcc56e...  ./intake/task3_large_items_check.txt
251e1d6ee34a0d99...  ./intake/task3_repo_items_check.txt
fd236a6a68c67e31...  ./live_response/00_collection_log.tsv
bf5d8f4fd74e6ec5...  ./live_response/01_clock.txt
ac477e92418469ec...  ./live_response/02_system.txt
f27e842b39dc254d...  ./live_response/03_processes.txt
373b38e0bc885714...  ./live_response/04_process_exe_sha256.txt
b7504b32ee4fe303...  ./live_response/05_open_files.txt
0fe81e4e60dab3c8...  ./live_response/06_logged_in.txt
2b33344eaa500370...  ./live_response/07_sockets.txt
a3e178e7b509d896...  ./live_response/08_listening.txt
4faebcf351180d0e...  ./live_response/09_interfaces.txt
a204d992a62c3743...  ./live_response/10_firewall.txt
f1d272f1b721bd55...  ./live_response/11_dns_hosts.txt
6c259643e1afd5fc...  ./live_response/12_kernel_modules.txt
9cdc73380dde6ffc...  ./live_response/13_kernel_log.txt
c60e63330200dc1a...  ./live_response/14_services.txt
15a14030a41165f5...  ./live_response/15_timers_cron.txt
511113ca2f128b86...  ./live_response/16_mounts_disks.txt
8d8e1cb55eca11de...  ./live_response/17_accounts.txt
57e69d2aaf4a5613...  ./live_response/18_sudoers.txt
7234b6aae85ed0c9...  ./live_response/19_ssh_keys.txt
d227418b4356b3c7...  ./live_response/20_lastlog.txt
dbc46ac7c74ee77c...  ./live_response/21_recent_files.txt
a87753adc849440f...  ./live_response/22_hidden_dirs.txt
15474608a0c119a8...  ./live_response/23_suid_sgid.txt
0adbcb6e9780d6f6...  ./live_response/24_shell_history.txt
36d7bd7809b8829b...  ./memory/uploader_region_search.txt
6be007d0ec50f487...  ./memory/vol_banners.txt
12d532df9ce7b433...  ./memory/vol_bash.txt
82fa2fe57ba49db5...  ./memory/vol_check_modules.txt
a88082cd04282f87...  ./memory/vol_check_syscall.txt
d440e21157a7da7e...  ./memory/vol_envars_uploader.txt
e65afb68c3047c60...  ./memory/vol_kmsg.txt
4c7f87384aac514f...  ./memory/vol_lsmod.txt
16ea59420cb8c2e5...  ./memory/vol_lsof_uploader.txt
91f8607db635d231...  ./memory/vol_malfind.txt
1832ded481995f6a...  ./memory/vol_maps_uploader.txt
21ee7ae04a82e34d...  ./memory/vol_psaux.txt
d68a63bce93b8ffb...  ./memory/vol_pslist.txt
7bacb0ed336b73fb...  ./memory/vol_pstree.txt
9242705cb12d0d09...  ./memory/vol_sockstat.txt
3a8371c7d684e6a0...  ./steps.csv
cd3e90d6a57ebeb0...  ./timeline/key_events.csv
39c62683e4df3885...  ./timeline/session_correlation.csv
e89ecb1f2047742b...  ./timeline/source_counts.csv
c94e5f64009a31bb...  ./timeline/supertimeline_IR1-2026-002.csv
37bfe3d3945def0d...  ./transcripts/00_workstation.txt
0b55bbe176e179cf...  ./transcripts/01_intake_verification.txt
8be6ad1200f6595a...  ./transcripts/02_live_response.txt
0543d44443883ac0...  ./transcripts/03_memory_analysis.txt
d3a0f8bc86e20bf0...  ./transcripts/04_disk_acquisition.txt
# outside git: /Users/Adult/ir1-vm-lab/evidence/IR1-2026-002
fa80be62ad1c0568...  IR1-2026-002_parrot-vm_seized.raw
fa80be62ad1c0568...  IR1-2026-002_vda.dd
```